SUNIL KULALI

AI Engineer | Generative AI | Machine Learning | LLMs

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PROFESSIONAL SUMMARY

AI Engineer with hands-on experience building LLM-powered applications, Retrieval-Augmented Generation (RAG) pipelines with LangChain and FAISS, and text-to-SQL analytics workflows with LangGraph, delivered through FastAPI REST APIs and PostgreSQL. Built supervised ML pipelines in Python and Scikit-learn during Machine Learning and Data Analyst internships, plus data engineering workflows on Azure. Seeking AI Engineer, Generative AI Engineer, or Junior ML Engineer roles.

TECHNICAL SKILLS

AI & Machine Learning: Python, Scikit-learn, XGBoost, Pandas, NumPy, Feature Engineering, Model Evaluation, Supervised Learning, Hyperparameter Tuning, MLflow, TensorFlow & PyTorch (foundational)

Generative AI & LLMs: LLMs, RAG, LangChain, LangGraph, Prompt Engineering, LLM Agents, Text-to-SQL, Embeddings, Semantic Search, Vector Databases, FAISS; Chroma, Pinecone, Weaviate (foundational)

Backend & Databases: FastAPI, REST APIs, JWT, RBAC, SQL (CTEs, Window Functions, Stored Procedures), PostgreSQL, MySQL, SQL Server, ETL/ELT, Data Processing, Next.js, React, Streamlit

Cloud & Deployment: Google Cloud Platform, Azure Data Factory, Azure Data Lake Storage, Databricks, PySpark, Docker (foundational), Git, GitHub, Testing, Debugging

Analytics & Visualization: Power BI, DAX, Power Query, Data Modeling, Data Visualization, EDA

EXPERIENCE

Machine Learning Intern | CODE-B Solutions Pvt. Ltd., Mumbai July 2025 – October 2025

- Developed supervised ML models (Random Forest, XGBoost, Logistic Regression), improving accuracy through feature engineering and hyperparameter tuning.
- Reduced model training errors by ~15% through outlier treatment, data preprocessing, and systematic testing.
- Wrote SQL queries, views, and stored procedures to prepare and retrieve data for model training.

Data Science Intern | IT Vedant, Bengaluru June 2024 – April 2025

- Automated ETL and API-driven reporting pipelines with Power BI and Power Query, improving reporting efficiency by 40% and data accuracy.
- Analyzed 10,000+ records with Python (Pandas, NumPy) and SQL for cleaning, validation, and EDA, delivering KPI-driven insights.
- Built Power BI dashboards (DAX, Power Query) tracking KPIs across business units; documented SOPs.

PROJECTS

InsightOS – Autonomous AI Business Intelligence & Decision Intelligence Platform

Technologies: *Python, FastAPI, Next.js, PostgreSQL, LangGraph, LLMs, RAG, Text-to-SQL, Google Cloud Platform*

- Built an LLM-powered BI platform that turns natural-language business questions into insights using text-to-SQL and RAG, with LangGraph workflows for SQL analytics, root-cause analysis, forecasting, recommendations, and automated reports.
- Implemented a multi-tenant architecture with role-based access control (RBAC), JWT authentication, and tenant-level data isolation on PostgreSQL.
- Added SQL validation, audit logging, AI observability, and human-in-the-loop approvals for safer, auditable AI-generated queries; integrated data-source connectors and Google Cloud services for ingestion and deployment.

DataIntel AI – Generative AI-Powered Data Analytics Assistant

Technologies: *Python, FastAPI, React, PostgreSQL, LangChain, FAISS, RAG, Power BI*

- Built a GenAI analytics assistant that answers business questions over structured data using a RAG pipeline with document processing, embeddings, FAISS vector search, and LangChain to ground LLM responses in retrieved context.
- Built a schema-agnostic KPI classification engine and a self-validation layer that recomputes KPIs and flags drift above 1%, backed by a 27-test backend regression suite.
- Developed FastAPI endpoints connecting AI services, databases, and the React frontend; added Power BI dashboards for KPI monitoring.

Healthcare Data Lake – Cloud-Based Data Engineering & Analytics Platform

Technologies: *Azure Data Factory, Azure Data Lake Storage, Databricks, PySpark, SQL, Power BI, MLflow*

- Designed a Bronze, Silver, and Gold medallion architecture to organize raw, cleansed, and analytics-ready healthcare data.
- Built ETL/ELT workflows with Azure Data Factory and PySpark for data ingestion, transformation, and data quality checks.
- Developed curated datasets and Power BI dashboards for healthcare KPI reporting; explored MLflow lifecycle management and AI-powered data retrieval.

Loan Approval Prediction & Credit Risk Analysis | Python, Scikit-learn, Random Forest, Streamlit

- Built an end-to-end ML pipeline on 45,000+ records (preprocessing, feature engineering, model comparison); across 5 classifiers, Random Forest reached 90.2% accuracy.
- Deployed the model as an interactive Streamlit web app serving real-time loan approval predictions.

EDUCATION

B.E. Computer Science & Engineering | Visvesvaraya Technological University | CGPA: 7.82 / 10 **2020 – 2024**

CERTIFICATIONS

- Master in Data Science & Analytics with AI — **I.T Vedant**
- Microsoft Certified: Fabric Data Engineer Associate (DP-700) — **Microsoft**
- Data Analytics & Visualization Job Simulation — **Accenture**
- Databases & SQL for Data Science with Python — **IBM**
- Machine Learning with Python — **IBM**